

C. Vladimir Rodríguez-Caballero

Department of Statistics, ITAM

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Updated: May 14, 2026

Research Fields

Time series econometrics (long memory, fractional integration, cointegration); high-dimensional factor models (multilevel dynamic factor models, FI-ML-DFM); scenario-based predictive densities (Growth-at-Risk, GiS/GaR); macroeconometrics and forecasting.

Education

Aarhus University / CREATES

Aarhus, Denmark

PhD in Econometrics

2013–2016

School of Business and Social Science; Center for Research in Econometric Analysis of Time Series (CREATES).

Dissertation: On factor analysis with long-range dependence. [\[Link\]](#)

Advisor: Niels Haldrup.

Universidad de Guanajuato

Guanajuato, Mexico

M.Sc. in Economics (Econometrics) — Summa Cum Laude

2009–2011

Dissertation: Ensayos sobre la Granger Causalidad. [\[Link\]](#)

Advisor: Daniel Ventosa-Santaulària.

Facultad de Ciencias, UNAM

Mexico City, Mexico

B.Sc. in Actuarial Science

2001–2005

Dissertation: Inferencia Bayesiana para la volatilidad en el modelo Black & Scholes. [\[Link\]](#)

Advisors: Alejandro Villagrán Hernández and Ramsés H. Mena Chávez.

Academic Appointments

Duke University, Department of Economics

North Carolina, USA

Visiting Professor

Aug 2025 – Jun 2026

ITAM, Department of Statistics

Mexico City, Mexico

Associate Professor (Tenured)

Aug 2023 – present

ITAM, Department of Statistics

Mexico City, Mexico

Assistant Professor (Tenure Track)

Aug 2018 – Jul 2023

Universidad Carlos III de Madrid, Department of Statistics

Madrid, Spain

Assistant Professor of Econometrics (Tenure Track)

Sep 2017 – Jul 2018

Universidad Carlos III de Madrid, Department of Statistics

Madrid, Spain

Postdoctoral Researcher

Jan 2017 – Aug 2017

Aarhus University / CREATES

Aarhus, Denmark

Research Assistant

Sep 2016 – Nov 2016

CREATES, Aarhus University / Danish National Research Foundation

Aarhus, Denmark

PhD Fellow

Sep 2013 – Aug 2016

Affiliations

CREATES, Aarhus University
Research Fellow

Aarhus, Denmark
Nov 2016 – Dec 2022

Professional Experience (Non-Academic)

HSBC Holding
Credit Risk Analyst / Credit Risk Manager
Risk Management.

Mexico City, Mexico
Feb 2006 – Jan 2009

Banco Fácil
Credit Risk Manager
Risk Management.

Mexico City, Mexico
Mar 2007 – Jan 2009

Microfinanciera Creciendo
Credit Risk Manager
Risk.

Mexico City, Mexico
Sep 2011 – Feb 2012

Santander Private Banking, Marketing Intelligence
Subdirector, Statistical Models and Methodology

Mexico City, Mexico
Apr 2012 – Mar 2013

External Research Collaborations

Interamerican Development Bank
Pollution and education.

2025

Interamerican Development Bank
Evaluación de la Fundación Camino a Casa.

2021–2022

Interamerican Development Bank
Gender equality and female labor participation.

2020

Interamerican Development Bank
Remittances payments through central banks.

2019

Articles [Quartiles: JCR, Scimago, Scopus]

- (24) Belloc, G., Garrón, I., Rodríguez-Caballero, C. V., & Ruiz, E. (2026). **FARS: Factor Augmented Regression Scenarios in R**. *Journal of Statistical Software*. [Forthcoming] [Q1, Q1, Q1]
- (23) Rodríguez-Caballero, C. V., & Ruiz, E. (2026). **Temperature in the Iberian Peninsula: Trend, seasonality, and heterogeneity**. *Annals of Applied Statistics*, 20(1), 173–196. [Q1, Q1, Q1] [\[Link\]](#)
- (22) Garrón, I., Rodríguez-Caballero, C. V., & Ruiz, E. (2026). **International factors and inflation risks**. *International Journal of Forecasting*, 42(3). [Q1, Q1, Q1] [\[Link\]](#)
- (21) Trillo, F. H., Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2025). **Monopoly unveiled: Telecom breakups in the US and Mexico**. *Economics Bulletin*, 45(4), 1989–1995. [~, Q3, Q3]
- (20) de Juan, A., Poncela, P., Rodríguez-Caballero, C. V., & Ruiz, E. (2025). **Economic activity and climate change**. *International Review of Environmental and Resource Economics*, 19(2). [Q3, Q3, Q3] [\[Link\]](#)
- (19) Gonzalez-Rivera, G., Rodríguez-Caballero, C. V., & Ruiz, E. (2024). **Expecting the unexpected: Stressed scenarios for economic growth**. *Journal of Applied Econometrics*, 39(5), 926–942. [Q1, Q1, Q1] [\[Link\]](#)
- (18) Caporin, M., Rodríguez-Caballero, C. V., & Ruiz, E. (2024). **The factor structure of exchange rates volatility: Global and intermittent factors**. *Empirical Economics*. [Q2, Q2, Q2] [\[Link\]](#)

- (17) López-Marmolejo, A., & Rodríguez-Caballero, C. V. (2023). **Assessing the effect of gender-related legal reforms on female labour participation and GDP per capita in the Central American region.** *Regional Statistics*, 13(3). [~, Q2, Q2] [\[Link\]](#)
- (16) Ergemen, Y. E., & Rodríguez-Caballero, C. V. (2023). **Estimation of a dynamic multi-level factor model with possible long-range dependence.** *International Journal of Forecasting*, 39(1), 405–430. [Q1, Q1, Q1] [\[Link\]](#)
- (15) Prados de la Escosura, L., & Rodríguez-Caballero, C. V. (2022). **War, pandemics, and modern economic growth in Europe.** *Explorations in Economic History*, 86, 101467. [Q1, Q1, Q1] [\[Link\]](#)
- (14) Rodríguez-Caballero, C. V., & Villanueva-Domínguez, M. (2022). **Predicting cryptocurrency crash dates.** *Empirical Economics*, 63, 2855–2873. [Q2, Q2, Q1] [\[Link\]](#)
- (13) Rodríguez-Caballero, C. V., & Vera-Valdés, J. E. (2021). **Air pollution and mobility: What carries COVID-19?** *Econometrics*, 9, 37. [Q2, Q2, Q2] [\[Link\]](#)
- (12) López-Marmolejo, A., Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2021). **Remittances at record highs in Latin America: Time to revisit the Dutch disease.** *Economics Bulletin*, 41(3), 2133–2146. [~, Q3, Q3] [\[Link\]](#)
- (11) Vera-Valdés, J. E., & Rodríguez-Caballero, C. V. (2021). **Air pollution and mobility in the Mexico City metropolitan area in times of COVID-19.** *Atmósfera*, 36(2), 343–354. [Q4, Q3, Q3] [\[Link\]](#)
- (10) Cataño, D. H., Rodríguez-Caballero, C. V., Chiann, C., & Peña, D. (2021). **Wavelet estimation for factor models with time-varying loadings.** *International Journal of Wavelets, Multiresolution and Information Processing*, 20(1), 2150033. [Q3, Q3, Q3] [\[Link\]](#)
- (9) Rodríguez-Caballero, C. V. (2021). **Energy consumption and GDP: A panel data analysis with multi-level cross-sectional dependence.** *Econometrics and Statistics*, 23, 128–146. [Q2, Q2, Q2] [\[Link\]](#)
- (8) Rodríguez-Caballero, C. V., & Vera-Valdés, J. E. (2020). **Long-lasting economic effects of pandemics: Evidence on growth and unemployment.** *Econometrics*, 8(3), 37. [Q2, Q2, Q2] [\[Link\]](#)
- (7) Rodríguez-Caballero, C. V., & Caporin, M. (2019). **A multilevel factor approach for the analysis of CDS commonality and risk contribution.** *Journal of International Financial Markets, Institutions and Money*, 63, 101144. [Q1, Q1, Q1] [\[Link\]](#)
- (6) Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2017). **Energy-growth long-term relationship under structural breaks: Evidence from Canada, 17 Latin American economies and the USA.** *Energy Economics*, 61, 121–134. [Q1, Q1, Q1] [\[Link\]](#)
- (5) Ergemen, Y. E., Haldrup, N., & Rodríguez-Caballero, C. V. (2016). **Common long-range dependence in a panel of hourly Nord Pool electricity prices and loads.** *Energy Economics*, 60, 79–96. [Q1, Q1, Q1] [\[Link\]](#)
- (4) Rodríguez-Caballero, C. V., & Knapik, O. (2014). **Bayesian log-periodic model for financial crashes.** *The European Physical Journal B*, 87(10), 1–14. [Q2, Q3, Q3] [\[Link\]](#)
- (3) Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2014). **Granger causality and unit roots.** *Journal of Statistical and Econometric Methods*, 3(1), 97–114. [\[Link\]](#)
- (2) Ventosa-Santaulària, D., & Rodríguez-Caballero, C. V. (2013). **Polynomial regressions and nonsense inference.** *Econometrics*, 1(3), 236–248. [\[Link\]](#)

- (1) Espin-García, O., & Rodríguez-Caballero, C. V. (2013). **Metodología para un scoring de clientes sin referencias crediticias**. *Cuadernos de Economía*, 32(59), 137–162. [Q3, Q4, Q3] [\[Link\]](#)

Books

- (1) Rodríguez-Caballero, C. V., & Espin García, O. (2012). **Matemáticas Financieras II**. Editorial GAFRA, Puebla, México. [\[Link\]](#)

Book Chapters

- (4) Rodríguez-Caballero, C. V., López Marmolejo, A., & Mendez Rolón, E. (2023). **Efectos del Modelo de Atención de la Fundación Camino a Casa**. In *Modelo de atención para recuperar y empoderar vidas: quince años uniendo esfuerzos*. Inter-American Development Bank. [\[Link\]](#)
- (3) Rodríguez-Caballero, C. V. (2023). **Approximation of hate detection processes in Spanish and other non-Anglo-Saxon languages**. In *News Media and Hate Speech Promotion in Mediterranean Countries* (pp. 65–80). IGI Global. [\[Link\]](#)
- (2) Rodríguez-Caballero, C. V. (s.f.). **Granger causalidad espuria en la relación de cartera total y vencida de créditos**. In *Volumen IV: Mercados, modelos financieros y entorno económico*. UAM.
- (1) Rodríguez-Caballero, C. V. (s.f.). **La inferencia bayesiana en la administración de riesgos**. In *Volumen II: Mercados, modelos financieros y entorno económico*, p. 181. UAM.

Technical Reports

- (2) Rodríguez-Caballero, C. V., López, A., & Ventosa-Santaulària, D. (2019). **Remittances payments through central banks: An application to the Central American countries exchange rates**. *IDB Technical Notes*, IDB-TN-1864. [\[Link\]](#)
- (1) Rodríguez-Caballero, C. V., & López, A. (2020). **Assessing the effect of gender equality before the law on female labor participation and GDP per capita in Central America, Panama, and the Dominican Republic**. *IDB Technical Notes*, IDB-TN-2128. [\[Link\]](#)

Pop-Science Articles

- (1) Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2025). **El impacto indeleble de la regresión**. *Miscelánea Matemática*, 81.

Software Development

FARS: Factor-Augmented Regression Scenarios (2025). R package for modelling and forecasting economic scenarios based on multi-level dynamic factor models. Available on CRAN: [\[Link\]](#)

Working Papers and Work in Progress

- (3) Rodríguez-Caballero, C. V., Bellocca, G. P., Garrón, I., & Ruiz, E. **The empirical distribution of sequential LS factors in multi-level dynamic factor models**. *R&R, Econometrics and Statistics*.
- (2) Rodríguez-Caballero, C. V., Pérez Castellano, J. L., & López Marmolejo, A. **Environmental inequality and educational outcomes: Evidence from air pollution in Mexico**. *Manuscript in preparation*.
- (1) Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. **Time Series Econometrics**. [Book in preparation]

Participation in Research Projects

1. Incertidumbre en modelos de factores dinámicos: Un índice de riesgo climático. Ministerio de Economía y Competitividad, PID2022-139614NB-C22, 2023–2027.
2. Climate change and economic challenges for the Spanish society. La Caixa Foundation, 2020–2021.
3. Indicadores económicos: Predicción con incertidumbre e inestabilidad. Ministerio de Economía y Competitividad, ECO2015-70331-C2-2-R.

Awards and Grants

Mexican National System of Researchers (SNI) — Secihti

SNI Level II

2024 – present

Mexican National System of Researchers (SNI) — Secihti

SNI Level I

2018 – 2023

ACRI and the Italian Econometric Association

Young Investigator Training Program (YITP) Prize

2016

International Association for Applied Econometrics

Travel Grant (competitive basis), 3rd IAAE Conference

2016

CREATES, Aarhus University / Danish National Research Foundation

Full PhD Fellowship

2013

Universidad de Guanajuato

Distinction: Best Academic Record, MSc 2009–2011

2011

CONACYT

Scholarship for research stay abroad

2011

Universidad de Guanajuato

Recognition of Academic Merit (2010 & 2011)

2011

CONACYT

CONACYT Graduate Scholarship (Master's)

2009–2011

Premio Nacional Francisco Aranda, Mexican Statistical Association

National Statistical Prize, 3rd place

2005

Best bachelor dissertation in statistics, academic years 2005–2006.

UNAM – Telmex Foundation

Excellence Scholarship

2003–2005

Teaching

ITAM: Undergraduate: Econometrics I, Regression Analysis, Time Series Econometrics, Introduction to Statistics.

Graduate (MSc): Financial Econometrics (Volatility), Time Series Econometrics, Advanced Econometrics (Asymptotic Theory), Climate Econometrics.

UC3M: Undergraduate: Statistical Methods for Social Sciences (Applied Time Series).

Graduate (MSc/PhD): Causal Inference.

Aarhus University: Quantitative and Limited Dependent Variables; Mathematical Economics II. (Teaching assistant)

CIDE (2013): MSc in Economics: Introduction to Multiple Time Series Analysis.

Supervised Students

BSc: A. Jørgensen & M. Sproegel (Aarhus, 2015); G. Santa Rosa García (2016); M. Cortés (2 theses, 2019); M. Villanueva (2 theses, 2020); P. Peralta (2022); R. Gómez (2023); P. Gracia (2023); P. Galeana (2023); L. Palacios (2024); X. Iñiguez (2024); A. Natividad (2024); G. Salazar (2025); A. L. García (2025); F. Pérez (2026).

MSc: M. Cañizares (2018); H. Frech (2018); M. Cortés (2022); R. Garduño (2023); P. Gracia (2026).

Research Stays

Universidad Carlos III de Madrid <i>Department of Statistics</i>	Madrid, Spain <i>Summer 2024</i>
Universidad Carlos III de Madrid <i>Department of Statistics</i>	Madrid, Spain <i>Summer 2022</i>
Universidad Carlos III de Madrid <i>Department of Social Sciences</i>	Madrid, Spain <i>Summer 2019</i>
Department of Statistical Sciences, University of Padova <i>Visiting PhD Scholar, ACRI YITP</i> Host: Massimiliano Caporin.	Padova, Italy <i>Spring 2016</i>
Department of Economics, Universidad Carlos III de Madrid <i>Visiting PhD Scholar</i> Host: Carlos Velasco.	Madrid, Spain <i>Spring 2016</i>
CIDE <i>Research visit</i>	Mexico City, Mexico <i>May – Jun 2013</i>
Institute of Mathematics, Cracow University of Technology <i>Research visit</i>	Cracow, Poland <i>Jan – Jul 2011</i>
Université Paris I (Panthéon-Sorbonne), SAMM Laboratoire <i>Research visit</i>	Paris, France <i>Jun 2011</i>

Seminars and Conferences (from 2015)

2025: Banco de Uruguay (Mar); Bank of England (Apr); Duke University (Oct); CIMAT Aniversario (Nov).

2024: 44th International Symposium on Forecasting, Dijon, France (Jul); 34th Midwest Econometrics Group Conference, Lexington KY, USA (Nov).

2023: Coloquio Mexicano de Economía Matemática y Econometría, Guanajuato (Sep); EcoSta 2023, Tokyo (Aug); Seminario ITAM; Seminario CIDE; Coloquio AME; Seminario CIMAT.

2022: Annual Conference IAAE, London; Arkansas State University–CQ; Regional Cooperation and Green Economy Conference, Liaoning University, China.

2021: Department Seminar, Örebro University, Sweden; 14th Workshop on Nonstationary Systems, Grodek, Poland; 26th International Panel Data Conference; XXXIV Foro Nacional de Estadística; CLATSE 2021; Universidad Veracruzana.

2020: International Symposium on Forecasting.

2019: Latin American Meeting of the Econometric Society, Puebla; Annual Conference IAAE, Cyprus; International Panel Data Conference, Vilnius.

2018: VIII Workshop in Time Series Econometrics, Zaragoza; Workshop on Econometrics and Data Science, Monterrey; Long Memory Conference, Aalborg; Semana de Matemáticas, ITAM; Department Seminar, Banco de México; Seminario Aleatorio, ITAM.

2017: Department of Statistics, UC3M, Madrid; 25th Annual Symposium SNDE, Paris; 40th International Panel Data Conference, Thessaloniki; 70th ESEM, Lisbon; Department Seminar, Universidad Pública de

Navarra.

2016: LNPP Seminar, CIDE, Mexico City; Department of Statistical Sciences, University of Padova; CREATES Anniversary Meeting, Sandbjerg; Annual Conference IAAE, Milan; Workshop in Time Series Econometrics, Zaragoza; PhD Workshop, Madrid.

2015: CREATES Lunch Seminar (Nov 2015 & Feb 2016), Aarhus; Annual DGPE Workshop, Sandbjerg.

Referee and Editorial Activities

Associate Editor: *Latin American Economic Review* (LAER), from October 2023.

Referee for: Trimestre Económico; Lecturas de Economía; Revista de Métodos Cuantitativos para la Economía y la Empresa; International Journal of Forecasting; The American Statistician; Latin American Economic Review; Journal of the Operational Research Society; Energy Economics; Empirical Economics; Journal of Nonparametric Statistics; Resources Policy; Energy Journal; Nature; Quantitative Finance; Journal of Econometrics; Regional Statistics; International Journal of Environmental Research and Public Health; Energies; Applied Economics Letters; Environment and Development Economics; Journal of Economic Structures; Atmosphere; Finance Research Letters; Communications in Statistics; Econometrics and Statistics; Financial Innovation; Communications in Statistics — Simulation and Computation; Revista de Matemática Teoría y Aplicaciones; Computational Statistics and Data Analysis.

Organization of Academic Events

- Department Seminar at ITAM. Principal organizer.
- Coloquio Virtual AME, 2020. Principal organizer.
- Mathematical and Statistical Methods for Actuarial Sciences and Finance (MAF 2018), Madrid. Member of the Scientific Committee.

Service in Selection Committees and Academic Societies

- Member, Comisión Dictaminadora 2, Área VI: Ciencias Sociales. Secihti.
- Premio Francisco Aranda Ordaz, AME, 2020–2021. Principal organizer.
- Academic evaluator for Secihti calls (PhD studies, postdoctoral stays, scientific projects).
- Member of multiple selection committees, Fulbright–García Robles program, COMEXUS.

References

Available upon request.